Environment, Health & Safety Plan

Preface / Purpose

PMC has generally had excellent safety track record. But still, we have had some accidents and incidents.

The objective of this manual is to act as a refresher course for old team and induction of the new team to Environment, Health & Safety (EHS).

Remember:

DO NOT, UNDER ANY CIRCUMSTANCES OR SCHEDULE PRESSURE, ACCEPT DEVIATION OF EHS MEASURES.

IF & WHEN AN ACCIDENT HAPPENS, NO ONE IS GOING TO LISTEN TO YOU THAT YOU ALLOWED EHS DEVIATION UNDER TIME PRESSURE OR CLIENT / CONTRACTOR PRESSURE.

Common EHS issues

Accidents and incidents causing injury, loss of life, material damage, etc. are a huge setback to any construction site.

In addition to loss of life & financial losses, they cause severe demoralisation of site teams, affect further activities and also cause bad publicity.

Even after proper induction and initial medical examination of workers and staff, continuous awareness and focus on EHS is required to ensure that accidents and incidents are avoided.

If we observe typical accidents on construction sites, they can be broadly categorised into following aspects:

Undiagnosed medical conditions of workers: Blood pressure, blood sugar, asthma, vertigo, allergies and colour blindness are the major causes of even well trained and aware workers and staff falling prey to accidents and causing accidents themselves.

General issues: Bad housekeeping, wrongly stored materials, unclean atmosphere, badly maintained movement paths, etc. Such issues can not only cause accidents but also can cause epidemic medical conditions for those working on sites.

Falling from height: People / material falling from heights above ground or into underground excavations, pits, etc.

Structural / staging collapse: Wrongly designed, erected or modified construction structures or staging collapse.

Soil / excavation collapse: Wrongly excavated soil / pits collapsing.

Electrocution: Electrical shocks through improper electrification and improper earth fault protection.

Fire, burns, explosions, glare damage, asphyxiation: Mishandling of fuels, inflammable gases, waste material and also mishandling of hot work equipment, not using welding shields, etc. *It should be noted that more fire deaths are due to asphyxiation (choking due to lack of oxygen) than due to fire itself.*

Asphyxiation / confined work spaces: Working and exposure to dangerous gases in confined work spaces without proper ventilation and safety gear.

Vehicle collision / toppling: Collision between vehicles, people, material, etc. due to improper movement, lack of traffic control, lack of reverse horns / signals, bad ground conditions, etc.

Hazardous chemicals: Mishandling and improper exposure to hazardous chemicals.

This document deals with how to analyse such risks and take preventive, and if required, corrective measures at all stages in the project.

The objective of this document / plan is not to set the limiting conditions for implementation of Environment, Health & Safety but to act as a guideline to further develop and improve the EHS conditions on the sites beyond the scope of this plan.

Should the clients need weekly / monthly reports, the same should be developed in consultation with their project team to suit their higher management reporting requirements.

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1. PMC policy & objectives of EHS Plan

Objective of EHS Plan

The purpose of this project EHS Plan is to describe how PMC will manage and control Environment, Health & Safety aspects of the project.

It is designed to enable those involved to enjoy an accident / incident free workplace and to prevent harm to people and environment at all stages i.e. design, tendering and construction / execution.

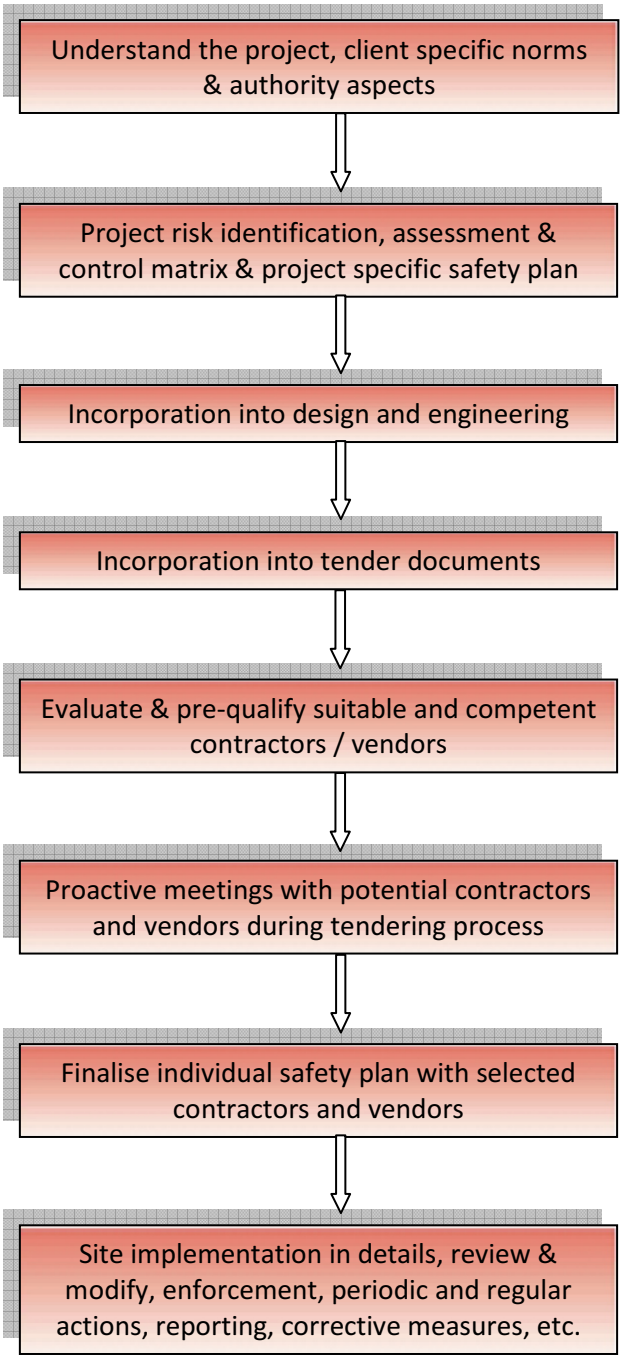
This plan is NOT an alternative to legislative compliance which should be maintained by all stakeholders with respect to their activities and responsibilities.

This EHS Plan is intended as a tool to support those requirements so as to:

- Identify and eliminate potential for significant injuries, incidents and occurrences
- Minimise occurrences of 'minor' accidents
- Maintain statutory compliance with respect to EHS
- Adhere to contractual obligations with client and ensure contractual obligations with various Contractors, vendors, etc.
- Encourage training, skilling, awareness and best practice
- Maintain accurate reporting and record keeping

2. Plan preparation procedure

This plan is to be developed in broad accordance with the flowchart given below:



3. Understanding the project, Client norms & Authority aspects

Project Safety Note

The Project / Site Manager should interact with the client as well as the designers (if on board).

He should also review client briefs / project briefs and client internal EHS norms, if available.

Having completed this, he should prepare a project safety note demonstrating his understanding of:

1. Nature of project including broad description, high risk / critical areas & constructions, etc.
2. Critical aspects in terms of buildings and forces of nature
3. Major EHS concern areas, hazards, etc.
4. Client expectations and norms, if any
5. Authority norms applicable for the project with specific deviations

Review and approval of Project Safety Note

This note should be reviewed by his immediate supervisor in PMC and approved.

PMC supervisor should give his constructive inputs based on previous experience. In case of high hazard or very critical projects, this note may be reviewed by suitable members of PMC CORE SENIOR MANAGEMNT TEAM.

This note forms the basis of development of project risk assessment matrix.

4. Project Risk identification, Assessment and Control

Various activities or aspects of the project can be classified into three categories as below based on the severity of the risk:

Severity	Description
P1	**Potential** Death, Permanent Disability or Major Structural Damage or Potential incident resulting in permanent or significant detrimental impact on natural or manmade elements.
P2	**Potential** Temporary Disability or Minor Structural Damage or Potential incident impacting natural or manmade elements that can be contained and reversed to acceptable conditions with no long term effect. or Any violation of statutory condition.
P3	**Potential** incident which have the potential to cause, minor damage to the environment or persons that would require first aid.

Project Risk Identification, Assessment & Control Matrix is attached as Annexure 2 of this document.

Purpose of the matrix:

- List each critical step / aspect that may adversely affect the project EHS.
- Identify the risks & severity associated with those steps / aspects.
- Present the preventive and remedial measures to mitigate or reduce the severity.

5. Design and Tendering phase

Design and engineering

- Prepare detailed list of various codes / norms to be followed for the project.
- Inform the same to the designers / engineers.
- Request for / conduct peer review, if applicable.
- Buildability check of drawings / specifications should also look into EHS aspects / safe construction.

Incorporation into tender packages

- Refer PMC standard document attached herewith.
- This document MUST form a part of tender document for each package / contract.

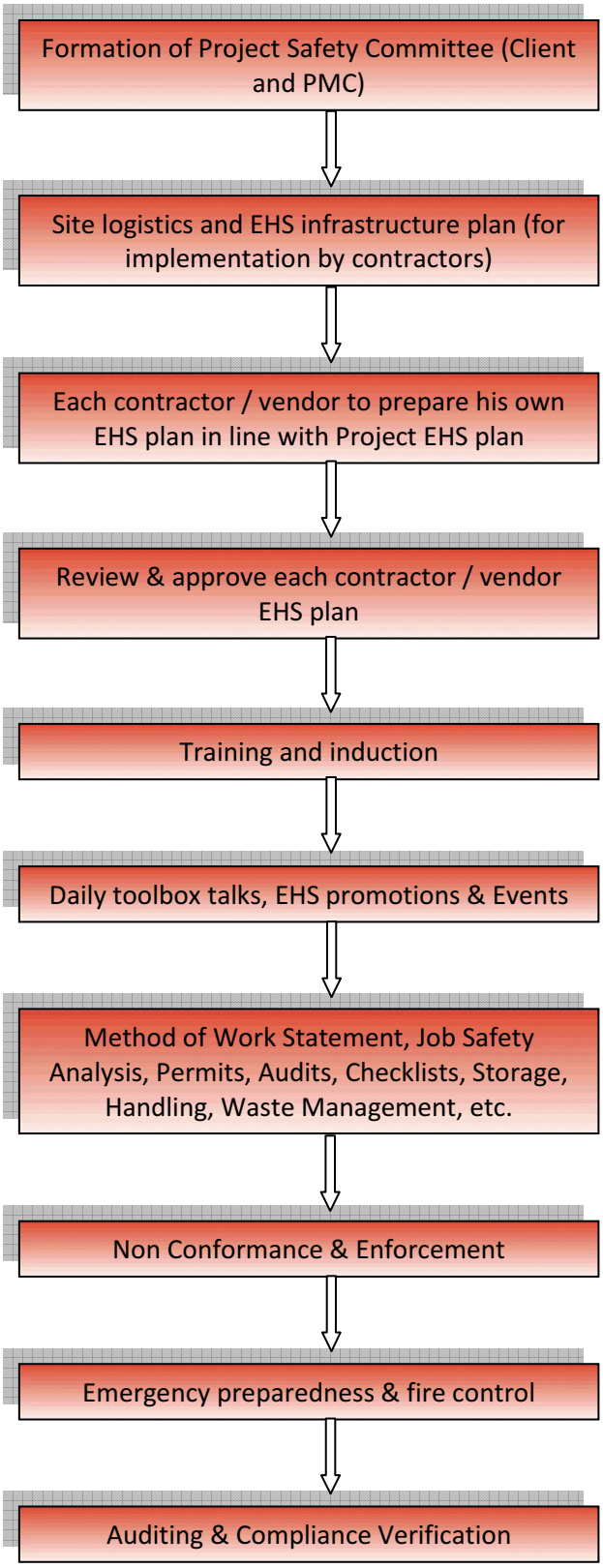
Prequalification of vendors / contractors

- Refer PMC standard PQ evaluation format attached herewith.
- Various vendors / contractors should be assessed for their EHS commitment, compliance and track record including:
    - Documentary compliance
    - Membership / awards of EHS organisations
    - Higher management commitment to EHS
    - Visits to sites
    - References from past customers
    - Similar projects executed

Pre and post-bid meetings

- Exclusive minuted meetings to be held at pre and post-bid stages to explain and emphasise EHS aspects.
- Explain each page of EHS document to the prospective contractors / vendors and get physical sign-off on all pages of the EHS document. This sign-off also helps in avoiding commercial claims for EHS measures implementations.

6. Construction / Execution phase



Project Safety Committee

This committee formally addresses and implements EHS measures.

- Initial committee comprises of: Project Manager, Site Safety Team, Client's Representative.
- Subsequently various contractors' / vendors' safety personnel are added to this.

Site logistics and EHS infrastructure plan (for implementation by contractors)

- Detailed site layout showing:
    - Final buildings & road layouts
    - *Segregated and without crisscross* PPE and non-PPE areas
    - Construction roads / paths
    - Men and material movement during construction with *minimal crisscross*
    - Various yards and storage areas
    - Temporary work areas
    - Safety park and training areas
    - First aid / medical facilities, ambulance parking
    - Facilities for labour (potable water, toilets, canteen, crèche)
    - Evacuation plan including emergency assembly points
    - Site / construction offices, vehicle parking facilities
    - Crèche / childcare centres
    - etc.
- No labour / workmen residential facilities at site.
- Ensure that site has male nurse as this proves to be a point of contention w.r.t. cultural issues.

Contractor / vendor EHS plan – Preparation, review and approval

- Share the Project EHS plan with each contractor / vendor on board.
- Each contractor / vendor to make individual EHS plan related to nature of contract / work, all the activities, scope, etc.
- Review, modify and approve contractor / vendor EHS plan.

Training and induction

- Detailed induction and training program to be formulated for all workmen, visitors, etc. including videos, etc.
- Initial and routine medical check-ups for all workers, engineers, etc. (Particularly health conditions like blood pressure, asthma, blood sugar, vertigo, colour blindness and allergies).

- All entrants to the site to be inducted by EHS Engineer / Officer.

- Visitors to be inducted, given PPE and accompanied by experienced staff to respective location, meeting or work area.

- Site team members (workers, supervisors, engineers) to be:

    - Given general induction
    - Given ob specific training
    - Given training in use of PPE
    - Subjected to pre-site brief medical examination
        - Blood pressure check
        - Colour blindness
        - Declaration of pre-known medical conditions, allergies and necessary measures

Toolbox talks, EHS Promotions & Events

- Daily toolbox talks to be given by contractor's safety officer / steward in presence of PMC / Client's safety engineer explaining that day's work to be carried out, precautions to be taken, increasing general safety awareness, etc.

- EHS Promotion & Events to be held including Safety Day, special recognition of workers and supervisors, sharing of experiences, etc.

- Prizes / bonuses to be given for encouraging adherence and awareness of EHS.

Method of Work Statement, Job Safety Analysis, Permits, Audits, Checklists, Storage, Handling, Waste Management, etc.

Various documentation formats and checklists have been given in the annexure for day-to-day activities, implementation, areas / specific issues to be checked, etc.

- **Method of Work Statement (MWS):** MWS is prepared by the respective trade contractor. This is reviewed and approved by PMC trade engineer. This document details the exact method / procedure in which the activity / job would be carried out.

- **Job Safety Analysis (JSA):** MWS is submitted to the Safety Team. This team understands and identifies the risks involved, precautions, mitigation measures, etc. which form the JSA.

- **Work Permits:** Based on training clearances of respective workmen & supervisors, MWS and JSA work permits are issued at site and work carried out accordingly.

- **Third party audits 1:** Third party audits and clearance for all heavy construction equipment (cranes, tower cranes, dumpers, tippers, trucks, dozers, material handling equipment, earthwork equipment, DG sets, etc.) is a must prior to entry into the site / being put to use (fixed / tower cranes). Any observed points should be closed and reaudit to be carried out till satisfactory approval.

- **Third party audits 2:** All critical / non standard staging (e.g. double height or more, overhangs / cantilevers, etc.) shall also be audited and approved by third party chartered structural engineer.

- **Checklists:** Various daily, weekly, monthly, etc. checklists are available for all regular equipment, tools, tackles, etc. used at typical construction site. These must be adhered to and implemented without exception. New items, if any, should be documented and approved by experts in the field in consultation with PMC senior management.

- **Storage and handling:** Detailed procedures for storage, handling, transport, etc. for various standard construction site items should be developed and implanted. These include steel, cement, fuels, pressurised gas cylinders, etc. also. These must be adhered to and implemented without exception.

- **Waste management:** Handling, storage and disposal of various types of waste generated on construction site has severe and serious impact on site conditions, environment, health, hygiene and safety. Detailed procedures for commonly encountered items should be developed and implemented based on site conditions and requirements.

Non Conformance & Enforcement

- In spite of best endeavours, there are non conformance issues arising at site. These should be dealt with and resolved expeditiously.

- Warning, re-warning and eviction system to be rigidly enforced so that lack of discipline and disregard to EHS issues do not go unnoticed and unpunished.

- Non Conformance Report (NCR) to be issued and linked to part payment of immediately forthcoming running account or stage or final payment bill (as the case may be) so that the contractors / vendors are forced to resolve the issues.

Emergency preparedness & fire control

- Site Emergency Response Team (ERT) to be formed.

- Detailed emergency response plan to be prepared involving various types of emergencies and responses to be given.

- Routine checks and test / mock drills to be carried out as per procedures laid out to check emergency preparedness and fire control.

- External auditors including local fire brigade to review and approve the drills.

- All fire fighting equipment, systems, trained staff, etc. to be reviewed on routine basis as per the procedures laid out.

Auditing & Compliance Verification

- PMC Regional Office (RO) and Head Office (HO) to conduct surprise / short-notice audits for EHS procedures compliance.

- Reports to be sent directly to PMC Managing Director and Operations Director with CC to respective Regional Manager and Project Manager.

- Immediate steps MUST be taken to carry out any mitigation measures as recommended by the audit report.

7. Annexures

01.00	Project Risk Identification, Assessment & Control Matrix
02.00	Contracts EHS Guidelines (Part of Contractor's Tenders)
03.00	Pre-Qualification
03.01	Pre-Qualification Questionnaire
03.02	PQ-Response Comparison
03.03	PQ-Ranking Evaluation
04.00	Project Safety Committee
05.00	Site logistics and EHS infrastructure plan
06.00	Induction & Training
06.01	Site safety rules & induction of regulars to the site
06.02	Standard safety presentation for visitors who need to enter hard hat (PPE) areas
07.00	Tool Box Talks – Standard Formats A to E
08.00	Method of Work statement
09.00	Job Safety Analysis
10.00	Work Permit
11.00	Pre Entry check list for all equipments, vehicles and machineries.
12.00	Daily Inspection check list for all equipments, vehicle and machineries
13.00	Daily Safety Observation report
14.00	Safety Violation (EHS NCR) report
15.00	Site instruction format
16.00	Near Miss Report Format
17.00	Accident Flash Report Format
18.00	Accident Investigation Report Format
19.00	Auditing and compliance verification